CHANDIGARH UNIVERSITY

NAAC GRADE A+
Accredited University

ASSIGNMENT

Subject: DBMS

Subject Code: 24CSH-298

Submitted By.
Name: Aryan Yadav
UID: 24BCF10079
Branch/Sec: 24AIT-KRG G2

Submitted To.
Mr. Shalabh Bhatiya
(Teacher's Name)

Assignment - 2

a i) Explain each type of database security issues

Database security issues refers to risks or problems that may lead to unauthorized access, misuse or loss of data. These issues can be classified in the following types.

i) Legal and Ethical Issues

Legal and Ethical concerns laws and regulations and moral responsibilities related to the storage and use of data in a database system. Organizations must ensure that they comply with data protection laws and privacy regulations. Sensitive information such as personal data, financial records and confidential business information should not be accessed and shared without proper authorization

ii) Policy issues

Policy issues related to the rules and guidelines established by an organization to protect its database systems. Organizations develop security policies that defines how data should be accessed, used and protected. These policies specify user roles access rights, responsibility of employees when handling data.

iii) System related Issues

System Related issues arise from technical weakness in database system or infrastructure like weak authentication system, software vulnerabilities improper configuration of database servers and lack of updates or security patches.

Q2) Explain the different database security control techniques.

Database security control techniques ~~are me~~ are methods used to protect database from unauthorized access and misuse.

i) Access Control

Access control restricts who can access the database and what operations they can perform, users are given specific permissions (read write, delete). Implemented using user accounts. Ensure that only authorized users can access sensitive data.

ii) Inference Control

Inference control prevents users from deducing sensitive information from accessible data.

iii) Flow Control

Flow control ensures that information does not flow from higher security levels to lower security levels.

It restricts how information moves within a system

iv) Data Encryption

It connects plain text data into coded form cipher text to protect it from unauthorized access.

Q3) Conside a table Employee (EmpID, EmpName, Salary, DeptID). Write an SQL query to create a view that displays only EMPID, EMPNAME, and DeptID from the Employee table

<u>SQL Query</u>

```
CREATE VIEW Employee_view AS
SELECT EmpID, EmpName, DeptID
FROM Employee
```

This view will display only EmpID, EmpName, and DeptID while hiding other columns such as salary.

Problem List

Submit

Premium

Description | Editorial | Solutions | Submissions

570. Managers with at Least 5 Direct Reports

Solved

Medium | Topics | Companies | Hint

SQL Schema > Pandas Schema >

Table: `Employee`

```
+-------------+---------+
| Column Name | Type    |
+-------------+---------+
| id          | int     |
| name        | varchar |
| department  | varchar |
| managerId   | int     |
+-------------+---------+
id is the primary key (column with
unique values) for this table.
Each row of this table indicates the
name of an employee, their department,
and the id of their manager.
```

1.8K | 367 | 66 Online

Code

MySQL | Auto

```mysql
# Write your MySQL query statement below
select b.name
from Employee a
left join Employee b
ON a.managerId = b.id
group by b.id
having count(b.id) >= 5
```

Saved | Ln 1, Col 1

Testcase | Test Result

Accepted

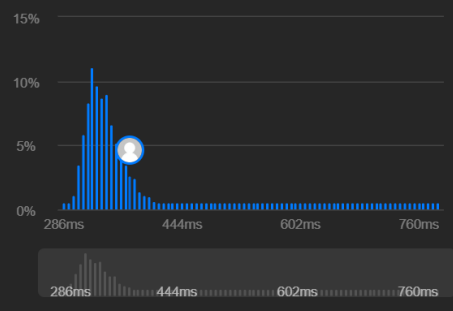
All Submissions

Accepted 11 / 11 testcases passed

aryanyadav_10079 submitted at Mar 25, 2026 10:28

Runtime

374 ms | Beats 21.29%



Code | MySQL

Problem List

Submit

Premium

Description | Editorial | Solutions | Submissions

577. Employee Bonus

Solved

Easy | Topics | Companies | Hint

SQL Schema > Pandas Schema >

Table: `Employee`

```
+-------------+---------+
| Column Name | Type    |
+-------------+---------+
| empId       | int     |
| name        | varchar |
| supervisor  | int     |
| salary      | int     |
+-------------+---------+
empId is the column with unique values
for this table.
Each row of this table indicates the
name and the ID of an employee in
addition to their salary and the id of
their manager.
```

1.6K | 223 | 40 Online

Code

MySQL | Auto

```mysql
# Write your MySQL query statement below
SELECT e.name, b.bonus
FROM Employee e
LEFT JOIN Bonus b
ON e.empId = b.empId
WHERE COALESCE(b.bonus, 0) < 1000;
```

Saved | Ln 6, Col 35

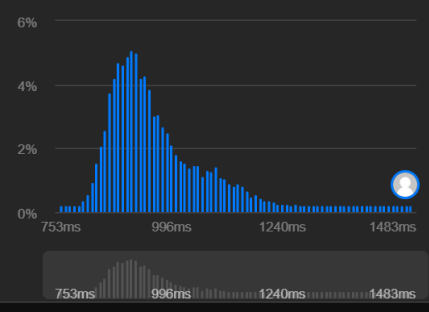
Testcase | Test Result

Accepted

All Submissions

Accepted 21 / 21 testcases passed

aryanyadav_10079 submitted at Mar 25, 2026 10:29

Runtime

4248 ms | Beats 5.01%